Digital Egypt Pioneers Initiative (DEPI)

Project Submission & Planning Document — October 16, 2026

Project Title: Railway Dataset Analysis & Interactive Power BI Dashboard

1. Project Planning & Management

Project Proposal & Overview

This project focuses on performing a comprehensive exploratory and statistical data analysis on the UK Railway Dataset using Python for data cleaning and preprocessing, followed by building an advanced, interactive Power BI dashboard. The core objectives are to analyze train punctuality, identify root causes of delays (e.g., signal failures, weather conditions), evaluate ticket sales patterns, and deliver actionable operational insights to improve railway service efficiency.

Team Structure & Task Assignment

Our team consists of 5 dedicated members, each assigned specific roles based on their core technical expertise:

- **Banan Magdy Mohamed (Team Leader & Data Cleaning Specialist):** Overall team leadership, task coordination, and leading the data preprocessing and cleaning pipeline (handling missing values, formatting dates, removing duplicates, and ensuring data integrity).
- **Hossam & Elham (Data Analysts):** Conducting Exploratory Data Analysis (EDA), statistical computations, identifying key trends, correlations, and performance metrics within the railway dataset.
- **Asmaa & Abdelrahman (Visualizations & Power BI Developers):** Designing, building, and deploying the interactive Power BI dashboards, crafting engaging visual data storytelling elements, and structuring user-friendly report layouts.

Project Timeline & Gantt Milestones

Item / Milestone	DEPI Industry Start Date	DEPI Final Submission Date
Project Planning & Management	2 Oct 2026	16 Oct 2026
Literature Review	9 Oct 2026	16 Oct 2026
Requirements Gathering	9 Oct 2026	16 Oct 2026
System Analysis & Design	9 Oct 2026	6 Nov 2026
Implementation (Source Code & Execution)	16 Oct 2026	30 Nov 2026
Final Presentation & Testing & Reports	16 Oct 2026	4 Dec 2026

* All project documentation, code repositories, and reports are fully documented and hosted on GitHub.

Risk Assessment & Mitigation Plan

- **Data Quality Issues (Missing or Corrupted Records):** Mitigated by executing rigorous data cleaning scripts under Banan's supervision to ensure a clean, reliable dataset prior to analysis and dashboard integration.
- **Schedule Delays or Integration Gaps:** Handled through strict internal milestones, collaborative reviews, and constant progress tracking among team members.

Key Performance Indicators (KPIs)

- 100% data cleaning accuracy and anomaly resolution.

- Comprehensive Power BI dashboard coverage including delay rates, revenue breakdowns, peak travel times, and station traffic analysis.
- Strict adherence to all DEPI submission deadlines and high professional standards.

2. Literature Review & Benchmarking

To establish a robust foundation for our project, our team conducted extensive research by reviewing similar business intelligence and transportation analytics repositories on **GitHub** and professional case studies shared on **LinkedIn**. By analyzing industry standards, benchmark dashboards, and open-source data pipelines:

- We identified best practices in data cleaning workflows and feature engineering for transportation datasets.
- We explored effective Power BI star-schema designs and interactive filtering techniques used in similar transit analytics projects.
- We integrated proven methodologies for visualizing punctuality metrics and delay attribution charts to maximize business value.

3. Requirements Gathering & Data-Driven Questions

Functional Requirements

- Database and dataset ingestion pipeline capable of handling transactional railway records.
- Dynamic filtering capabilities by route, station, ticket class, purchase type, and journey date.
- Automated calculation of delay metrics, refund requests, and punctuality percentages.

Non-Functional Requirements

- High query performance and smooth responsiveness across all Power BI report pages.
- Clean, professional UI/UX design adhering to modern data visualization principles.
- Maintainable, well-documented Python code and version-controlled GitHub repository.

Core Data-Driven Questions & Analytical Requirements

To ensure our analysis and Power BI dashboards address the core business objectives, the requirements gathering phase established the following key analytical questions that the dataset must answer:

- **Punctuality & Delay Analysis:** What is the overall percentage of delayed trains versus on-time arrivals, and what are the primary root causes of delays (e.g., signal failures, technical issues, weather conditions)?
- **Route & Station Performance:** Which departure and arrival routes experience the highest traffic volume and the most frequent delays?
- **Ticket Sales & Passenger Behavior:** How do ticket classes (Standard vs. First Class), purchase types (Online vs. Station counter), and payment methods (Credit Card, Contactless, etc.) impact sales volume and refund requests?
- **Financial & Operational Impact:** What is the total revenue generated across different routes, and how do delays correlate with refund claims and customer compensation trends?